

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 28

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. The isoefficiency function characterizes the rate at which the problem size (W) must increase relative to the number of processors (p) to maintain a constant efficiency. Derive the isoefficiency metric that defines this relationship. Hint: Start from $E = T_1/(pT_p)$, model $T_p = W/p + T_o(p)$, substitute, simplify to $E = W/(W + pT_o(p))$, then solve for W assuming E is constant to see how problem size must scale with p .
2. A parallel algorithm has a sequential work of W (number of operations). Its total communication overhead is determined to be $T_o = p \sqrt{n}$, where n is the problem size and p is the number of processors. Derive the isoefficiency function for this algorithm. Categorize this algorithm's scalability (e.g., highly scalable, scalable, or poorly scalable). If you double the number of processors, by what factor must you increase the problem size n to maintain constant efficiency?
3. Consider the unordered sequence: [8, 3, 4, 7, 9, 2, 1, 5].
 - A. Sort the integers into a single increasing list using the Bitonic Sorting phases.
 - B. Calculate the exact number of total compare-and-exchange steps required to sort these 8 numbers into a single increasing sequence.
 - C. If the sequence is currently bitonic, describe the state of the two resulting sequences after the first four compare-and-exchange operations are performed.

• Welcome: Sean

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